Problem Sheet #1

Author: Rosh Guadiana **Course:** Measure Theory
TA: Ronny Morales **Date:** 2026-03-31

– **Problem 1:**
If A and B are sets, show that $A \subseteq B$ if and only if $A \cap B = A$.

—— ***Solution:***
We proceed by mutual implication.

Forward Direction ($\Longrightarrow$): Assume $A \subseteq B$. We must show $A \cap B = A$. First, let $x \in A \cap B$. By the definition of intersection, $x \in A$ and $x \in B$. Thus, $x \in A$, which implies $A \cap B \subseteq A$. Second, let $x \in A$. Since $A \subseteq B$, it follows that $x \in B$. Because $x \in A$ and $x \in B$, we have $x \in A \cap B$. Thus, $A \subseteq A \cap B$. By mutual inclusion, $A \cap B = A$.

Reverse Direction ($\Longleftarrow$): Assume $A \cap B = A$. We know by that $A \cap B \subseteq B$, using our assumption we conclude that $A \subseteq B$

Q.E.D.

– **Problem 2:**
If A is a set and $\mathcal{B} = \{B_i : i \in I\}$ is any family of sets, show that:

$$A \cap \bigcup_{i \in I} B_i = \bigcup_{i \in I}(A \cap B_i) \text{ and } A \cup \bigcap_{i \in I} B_i = \bigcap_{i \in I}(A \cup B_i)$$

—— ***Solution:***
First part.

Forward Direction ($\Longrightarrow$): Let $x \in \left(A \cap \bigcup_{i \in I} B_i\right)$.
By definition of intersection we have $x \in A \land \exists j \in I$ such that $x \in B_j$, so we have that $x \in A \cap B_j$, if this element is in the intersection of a sole set of the family, is also true that is in the union of all sets in the family, therefore $x \in \bigcup_{i \in I}(A \cap B_i)$

Reverse Direction ($\Longleftarrow$): Let $x \in \bigcup_{i \in I}(A \cap B_i)$
We know that $\exists j \in I$ such that $x \in B_j \cap A$, this is $x \in A \land x \in B_j$, because x is in any set of the family of sets, is also true that is in the union of the family, therefore $x \in \bigcup_{i \in I} B_i \land x \in A$, so we have that $x \in A \cap \bigcup_{i \in I} B_i$

Second part.

Forward Direction ($\Longrightarrow$): Let $x \in A \cup \bigcap_{i \in I} B_i$
So $x \in A \ \lor \ \forall j \in I,\ x \in B_j$ this is $x \in A \cup B_j$, for this we know that $\forall j \in I,\ x \in A \cup B_j$, this is $\bigcap_{i \in I}(A \cup B_i)$

Reverse Direction ($\Longleftarrow$): Let $x \in \bigcap_{i\in I}(A \cup B_i)$
so $\forall j \in I,\ x \in A \cup B_j$, so if $\forall j \in I,\ x \in A \vee x \in B_j$,by Law of excluded middle we divide this in two cases:

1. $x \notin A$
   - If this is the case, then $x \in B_j, \forall j \in I$ must be true, but this is trivially $x \in A \cup \bigcap_{i\in I} B_i$
2. $x \in A$
   - This is trivially $x \in A \vee \forall j \in I,\ x \in B_j$ therefore $x \in A \cup \bigcap_{i\in I} B_i$

Q.E.D.

– **Problem 3:**
Let B be any set. Show that $(A \cap B) \cup C = A \cap (B \cup C) \Longleftrightarrow C \subseteq A$.

—— ***Solution:***
Forward Direction ($\Longrightarrow$): Assume $(A \cap B) \cup C = A \cap (B \cup C)$.

Let $x \in C$, by the definition of the union it follows that $x \in (A \cap B) \cup C$, we know by assumption that this is equall to $x \in A \cap (B \cup C)$, so $x \in A \wedge x \in (B \cup C)$, for this expresion hold true we strictly require $x \in A$, therefore $x \in C \Rightarrow x \in A$ thus $C \subseteq A$

Reverse Direction ($\Longleftarrow$): Assume $C \subseteq A$.

- **Forward inclusion**: $(A \cap B) \cup C \subseteq A \cap (B \cup C)$
  - Let $x \in (A \cap B) \cup C$, this is $x \in A \cap B \vee x \in C$.
    - Case 1: $x \in A \cap B$, we have that $x \in A \wedge x \in B$, and for definition of union it holds that $x \in B \cup C$, therefore $x \in A \cap (B \cup C)$
    - Case 2: $x \in C$, for our assumption we know that this implies $x \in A$, also we know for definition of union $x \in B \cup C$, so $x \in A \wedge x \in B \cup C$, therefore $x \in A \cap (B \cup C)$
- **Backward inclusion**: $(A \cap B) \cup C \supseteq A \cap (B \cup C)$
  - Let $x \in A \cap (B \cup C)$, this is $x \in A \wedge x \in B \cup C$
    - Case 1: $x \in B$, we know that $x \in A$, this means $x \in A \cap B$, and for the definition of union $x \in (A \cap B) \cup C$ holds true
    - Case 2: $x \notin B$, therefore $x \in C$, so by definition of union $x \in (A \cap B) \cup C$

Q.E.D.

– **Problem 4:**
The symmetric difference of two sets A and B is the set D of all elements that belong to either A or B but not both, symbolically,

$$D \equiv A \Delta B = (A \cup B) - (A \cap B)$$

Represent D with a diagram and show that $D = (A - B) \cup (B - A)$.
Furthermore, show that
i) $A \subseteq A\Delta B \cup A$,
ii) $A \subseteq A\Delta B \cup B$,

iii) $B \subseteq A\Delta B \cup A$,
iv) $B \subseteq A\Delta B \cup B$.

— ***Solution:***
Show that $D = (A - B) \cup (B - A)$:
Let $x \in D$, this is $x \in (A \cup B) - (A \cap B)$, so we have

$$x \in A \cup B \wedge x \notin A \cap B \Longrightarrow x \in A \cup B \wedge x \in A^c \cup B^c$$

this is

$$(x \in A \wedge x \in A^c) \vee (x \in B \wedge x \in A^c) \bigwedge (x \in A \wedge x \in B^c) \vee (x \in B \wedge x \in B^c)$$

In the right side
$(x \in A \wedge x \in A^c) \vee (x \in B \wedge x \in A^c) \iff \text{FALSE} \vee (x \in B \wedge x \in A^c) \iff x \in B \wedge x \in A^c$
$\iff x \in B \wedge x \notin A \iff x \in (B - A)$
In the left side
$(x \in A \wedge x \in B^c) \vee (x \in B \wedge x \in B^c) \iff (x \in A \wedge x \in B^c) \vee \text{FALSE} \iff x \in A \wedge x \in B^c$
$\iff x \in A \wedge x \notin B \iff x \in (A - B)$
joining both sides we got

$$(A - B) \cup (B - A)$$

i) $A \subseteq A\Delta B \cup A$,
Let $x \in A$, for the definition of union $x \in A\Delta B \cup A$
ii) $A \subseteq A\Delta B \cup B$,
Let $x \in A$, thus $x \in A \cup B$ by definition of union. We have two cases.
Case 1: $x \in B$ therefore $x \in A\Delta B \cup B$, by the definition of union.
Case 2: $x \notin B$, we can't go directly as first case, instead we denote $x \in A \wedge x \notin B \iff x \in A - B$ therefore by the first demonstration we got that $x \in (A - B) \cup (B - A) \iff x \in A\Delta B \Longrightarrow x \in A\Delta B \cup B$
iii) $B \subseteq A\Delta B \cup A$,
We procceed like in ii), let $x \in B$
iv) $B \subseteq A\Delta B \cup B$.
We procceed like in i) letting $x \in B$

Q.E.D.

– Problem 5:
Show that if $f : A \to B$ and $E, F \subseteq A$ then $f(E \cup F) = f(E) \cup f(F)$ and $f(E \cap F) \subseteq f(E) \cap f(F)$.

—— ***Solution:***
First part:

Forward Direction ($\Longrightarrow$): Let $y \in f(E \cup F)$, by the definition of the function we have $\exists x \in E \cup F$ such that $f(x) = y$ thus $x \in E \vee x \in F$,
Case 1: $x \in E \Rightarrow f(x) \in f(E) \iff y \in f(E)$
Case 2: $x \in F \Rightarrow f(x) \in f(F) \iff y \in f(F)$
For this we have that $y \in f(E) \vee y \in f(F) \Longrightarrow y \in f(E) \cup f(F)$ thus $f(E \cup F) \subseteq f(E) \cup f(F)$

Backward Direction ($\Longleftarrow$): Let $y \in f(E) \cup f(F) \iff y \in f(E) \vee y \in f(F)$ thus:
Case 1:

Let $y \in f(E) \Rightarrow \exists x_E \in E$ such that $f(x_E) = y$ by definition of union $x_E \in E \cup F$ therefore $f(x_E) \in f(E \cup F) \iff y \in f(E \cup F)$
Case 2:

Let $y \in f(F)$ $\exists x_F \in F$ such that $f(x_F) = y \Longrightarrow x_F \in E \cup F$ therefore $f(x_F) \in f(E \cup F) \iff y \in f(E \cup F)$
so we know that $f(E) \cup f(F) \subseteq f(E \cup F)$
So we demonstrate that $f(E \cup F) = f(E) \cup f(F)$

Second part:

Let $y \in f(E \cap F)$ for definition of function $\exists x \in E \cap F$ such that $f(x) = y$ thus $x \in E \wedge x \in F$

Q.E.D.